



Temporal and vertical persistence of the non-native rotifer *Kellicottia bostoniensis* (Rousselet, 1908) during water stratification and mixing of a deep tropical reservoir

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Abstract The non-native rotifer *Kellicottia bostoniensis* (Rousselet, 1908) has been recorded for ten years in an oligotrophic monomictic deep tropical reservoir in southeastern Brazil, where this study was conducted. This study evaluated how water column conditions in this reservoir influence the occurrence and density of *K. bostoniensis* throughout the year. The occurrence and density of *K. bostoniensis*, phytoplankton composition, and physical and chemical variables were assessed from the surface to 40 m depth at the dam area for two consecutive years. The effects of seasonal stratification and dry-season

mixing were analyzed using nonparametric tests and linear mixed-effects models (LMM). *K. bostoniensis* occurred throughout the water column, with significantly higher densities during the dry seasons. Nanophytoplankton was positively associated with *K. bostoniensis* density, whereas cyanobacteria showed a negative relationship, indicating that food composition and size structure, rather than total algal biomass, is a key driver of population dynamics. *K. bostoniensis* exhibited higher densities at intermediate depths (5–25 m), and its vertical distribution was positively associated with the dissolved oxygen levels. Overall, our findings advance the understanding of the ecological drivers underlying the distribution of *K. bostoniensis*, highlighting how this non-native species responds to interacting vertical gradients and seasonal dynamics in tropical reservoir systems.

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Introduction

First described from samples collected in 1907 from a lake in Boston, USA, the rotifer species *Kellicottia bostoniensis* (Rousselet, 1908) is frequently cited in the literature as a non-native species and as potentially harmful whenever it is recorded in new locations. Commonly found in North America (Edmondson, 1959), the first record outside its native range occurred in 1943, when the species was reported in Sweden (Carlin, 1943). Since then, the species has been recorded worldwide and is frequently found across different countries and continents (Yang & Min, 2020; Villabona-González & López-Cardona, 2022; Nandini et al., 2022; Sługocki & Czerniawski, 2025). The main hypotheses for the species' intercontinental dispersal are via ballast water (Strayer, 2010) and passive transport of resting eggs by boat traffic, waterfowl, fish gills, or by migratory birds (Lopes & Lansac-Tôha, 1997; Lopes et al., 2025).

In South America, *K. bostoniensis* was first reported in 1997 in the Segredo Reservoir, located in the Iguaçú River basin in Southeast Brazil (Lopes & Lansac-Tôha, 1997), and subsequently in Argentina (José de Paggi, 2002). Currently, the species has been reported to be spreading toward the northern border of the Amazon region (Gomes et al., 2022). In Brazil, where the hydrographic basins have been widely used for energy generation through the construction of hydroelectric reservoirs, *K. bostoniensis* has been frequently detected in these systems in the south and southeast of the country (Bezerra-Neto et al., 2004; De-Carli et al., 2018; Macêdo et al., 2020; Lopes et al., 2025). In those reservoirs, *K. bostoniensis* has been found across varying trophic levels, with higher densities in oligotrophic conditions (Macêdo et al., 2020).

Kellicottia bostoniensis is recognized for its distinct morphology, presenting a transparent and conical body, a long spine in the anterior region and one in the posterior region, as well as three smaller spines in the anterior region, which gives it a unique appearance and high fluctuation potential (De-Carli et al., 2018; Zhdanova et al., 2019). In addition, Branco et al. (2023) found evidence of the growth of these spines as a form of defense against predation and for the establishment of the species in a tropical reservoir. Other studies indicate that *K. bostoniensis* is highly adaptable to variable trophic conditions,

being considered a euryecious species, which makes it well adapted for dispersal and establishment (José de Paggi, 2002; Gomes et al., 2022; Ferreira-Júnior et al., 2023). Furthermore, *K. bostoniensis* is characterized by having a high reproductive capacity (Lopes et al., 2025) and with an opportunistic microfilter-feeding habit on small particles, including microalgae, bacteria, and detritus (Gilbert, 2022). These attributes are often associated with highly invasive potential organisms, as they favor the dispersal and establishment of species in various freshwater environments, such as reservoirs, lakes, and rivers (Villabona-González & López-Cardona, 2022; Hong et al., 2025).

Invasive species are those that expand beyond their native range, primarily through human-mediated processes, and exert negative effects on local biodiversity, community organization, and ecosystem functioning (Ricciardi, 2007; Lockwood et al., 2013; Simberloff et al., 2013). Although Zolotareva et al. (2021) included *K. bostoniensis* as a transcontinental invasive species, recent studies have considered its impact to be minimal (Hong et al., 2025) or suggest that it lacks clear invasive characteristics (Arcifa et al., 2020). In the absence of obvious evidence of impact, the presence of the species outside its native range may reflect an expansion of its geographic distribution (Blackburn et al., 2011; Pardo et al., 2013; Daly et al., 2023; Pfadenhauer et al., 2024). However, the ecological effects of non-native species are often context-dependent, varying across ecosystems, environmental gradients, and disturbance regimes, potentially hindering the detection in short-term studies or restricted areas (Searcy et al., 2023; Potgieter et al., 2024). In some cases, non-native taxa may persist at low population levels or exert subtle and indirect influences on ecological processes without causing immediate or visible changes in community structure, a phenomenon known as silent invasions (Carneiro et al., 2025). Such effects may intensify over time or emerge under specific environmental conditions, especially in zooplanktonic organisms that exhibit high sensitivity and rapid responses to environmental changes, such as eutrophication, global warming, and aquatic habitat modifications (Miglietta & Lessios, 2009; Ricciardi & MacIsaac, 2011; Branco et al., 2023; Haubrock et al., 2025).

In this context, reservoirs are often identified as vulnerable to biological invasions and recognized

as favorable environments for the establishment of non-native species in freshwater ecosystems (Strayer, 2010; Havel et al., 2005), due to their physical and chemical characteristics, as well as high levels of environmental disturbance, making them more susceptible to invasions than natural lakes (Havel et al., 2005). In addition, these ecosystems can function as “stepping stones” for the spread of invaders through interconnected systems, facilitating their introduction into new environments (Pereira et al., 2018; Macêdo et al., 2020). Furthermore, deeper areas of large reservoirs, such as dam zones, may exhibit distinct features along the vertical profile of the water column, creating heterogeneous environments. Thus, vertical profiling of these systems reveals variations in the physical and chemical conditions, including temperature, dissolved oxygen, salinity, nutrients, and the distribution of aquatic organisms at different depths (Chang et al., 2022). The vertical analyses of some of these physical and chemical variables can reveal important environmental changes, such as thermal stratification and the formation of hypoxic zones, which can influence the distribution of organisms in the water column (Kalf, 2002; Dodds & Whiles, 2010; Ding et al., 2026).

The dynamics of thermal stratification and water-column mixing processes control the vertical distribution of nutrients and dissolved oxygen, playing a central role in the organization of planktonic communities in lakes and reservoirs (Wetzel, 2001). During stratification, the water column forms distinct layers (epilimnion, metalimnion, and hypolimnion) with different physical and chemical conditions, restricting vertical mixing of water and dissolved gases (Guarino et al., 2005; Branco et al., 2009). Additionally, the vertical distribution of producers and consumers is directly influenced by light availability, water-column stratification, and nutrient variability. The mixing of the water column, which can mean destratification of a body of water that has been stratified for a long period, can cause nutrients to rise from the hypolimnion to the epilimnion (Guarino et al., 2005), increasing phytoplankton productivity and influencing the entire plankton populations. These conditions generate different distribution zooplankton patterns along vertical profiles of the water column (Kessler & Lampert, 2018; Leach et al., 2019). In tropical reservoirs, seasonal shifts in rainfall regimes can directly influence thermal stratification, water-column stability, and nutrient availability, thereby

shaping planktonic community structure (Tundisi et al., 2010; Elmoor-Loureiro et al., 2023). For rotifers, these environmental variations may result in changes in population density and vertical distribution since they are often concentrated on surface layers or mixing zones, where food availability and oxygen conditions are more favorable (Marshall et al., 2024). Beyond this, dissolved oxygen acts as a physiological constraint for zooplankton, regulating respiration, survival, and vertical habitat use, particularly in stratified systems where strong oxygen gradients limit the distribution to deeper layers (Doubek et al., 2018; Jane et al., 2021). Thus, stratification and mixing processes act as key environmental filters that modulate the structure and dynamics of zooplankton communities along the water column.

This study aimed to investigate how water-column conditions in a deep tropical reservoir influence the population ecology of the non-native rotifer *Kellicottia bostoniensis*. We studied the occurrence and density of *Kellicottia bostoniensis* along the vertical profile in the dam zone of the Ribeirão das Lajes reservoir, where this species has been found since 2001 and attain high surface density in winter (Branco et al., 2023). We further assessed the relationship of the *K. bostoniensis* population with physical and chemical variables and phytoplankton groups. We tested the following hypotheses:

- (1) *Kellicottia bostoniensis* density is influenced by the seasonality, with higher values at all depths during the dry season in winter when the reservoir mixing process occurs, causing greater nutrient availability.
- (2) The increase of *K. bostoniensis* density during the mixing is directly associated with an increase in nano-phytoplankton density ($< 20 \mu\text{m}$);
- (3) As an euryecious species, *K. bostoniensis* occurs throughout the water column but preferentially reaches higher densities at higher dissolved oxygen concentrations.

Methodology

Study area

The Ribeirão das Lajes tropical reservoir (Fig. 1) is located between the municipalities of Pirai and Rio

Claro, in the Rio de Janeiro State, Brazil. Inaugurated in 1908, the reservoir is fed by the Piraí River and by local streams surrounded by Atlantic Forest remnants. The dam was constructed as one of the largest hydroelectric facilities in the country, supplying energy to the domestic and industrial sectors, and providing drinking water to neighboring municipalities (Vaz, 2012). The Ribeirão das Lajes reservoir has a surface area ranging from 30.7 km² to 47.8 km² and a mean residence time of 300 days, with a mean depth of 15 m (maximum depth of 40 m) and a volume of 4.5×10^8 m³ (Guarino et al., 2005; Lopes et al., 2017; Fintelman-Oliveira et al., 2022). The reservoir is an oligo-mesotrophic system with thermal stratification for most of the year and is classified as monomictic, i.e., a total or partial mixing occurs during the winter months (Macêdo et al., 2018; Klippel et al., 2020).

Sampling and sample analyses

A total of 38 sampling campaigns were conducted over 26 months, from November 2004 and December 2006, at the same sampling location in the dam zone of the Ribeirão das Lajes Reservoir (22°42'07.87" S, 43°53'01.46" W). Sampling was conducted monthly, with additional fortnightly campaigns during selected periods of stronger thermal stratification and potential water-column mixing (Table 1). According to the regional climatic regime, the rainy season occurs between November and April and the dry season between May and October.

Zooplankton samples were collected independently at 5 m depth intervals throughout the water column, from the surface to the reservoir bottom (35–40 m), using a Van Dorn bottle. At each sampling depth, 10 L of water was collected and filtered through a 68-µm mesh net. Thus, each depth represented an independent sample for subsequent analyses.

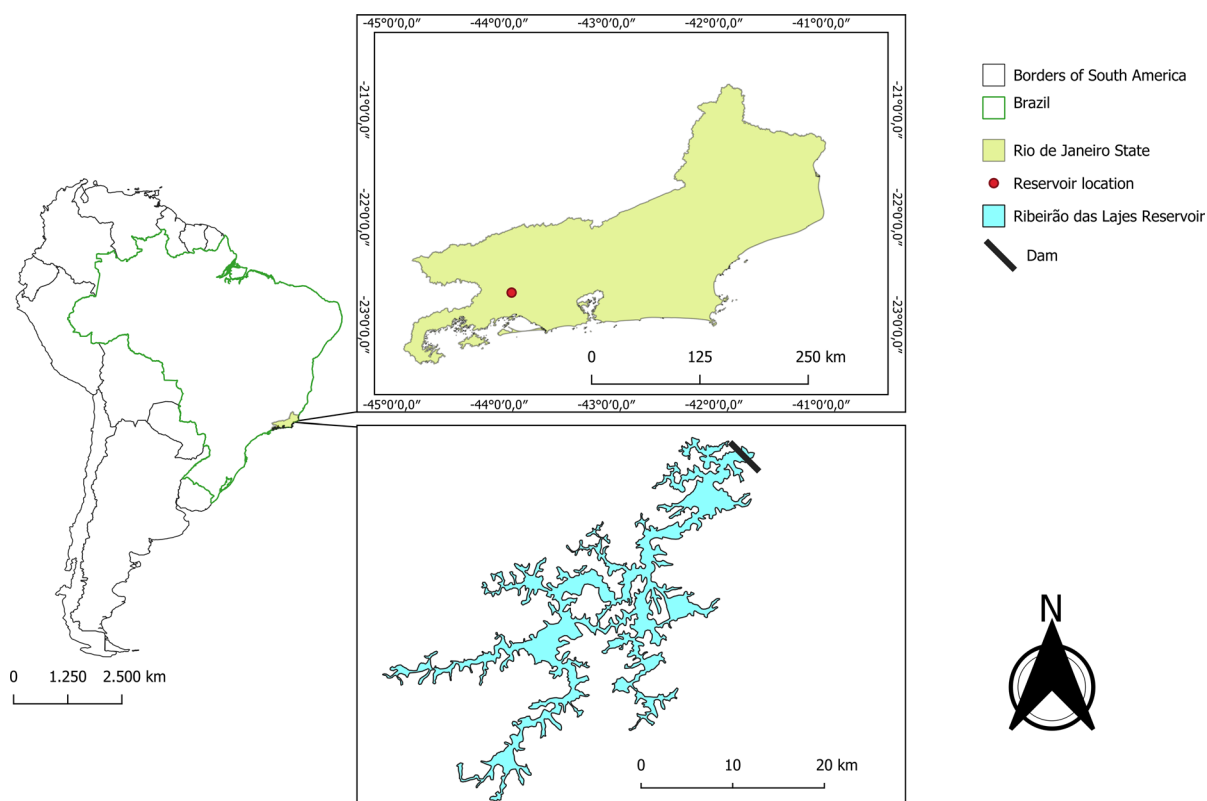


Fig. 1 The Ribeirão das Lajes Reservoir is located between the municipalities of Piraí and Rio Claro in the south part of the Rio de Janeiro State, in the Southeast Brazil

Table 1 Chronological summary of the sampling campaigns conducted in Ribeirão das Lajes Reservoir (November 2004–December 2006), including the sampling dates, number ofdepths sampled in each campaign, seasonal period (rainy or dry), and the availability of *Kellicottia bostoniensis* and phytoplankton data used in the analyses

Month/year	Campaign(s) day(s)	Sampled depths	Season	Zooplankton (<i>Kellicottia bostoniensis</i>) sampled?	Phytoplankton sampled?
Nov/04	16/nov	8 (0–35 m)	Rainy	Yes	Yes
	02/dec	8 (0–35 m)	Rainy	Yes	Yes
Dec/04	14/dec	8 (0–35 m)	Rainy	Yes	Yes
	29/dec	9 (0–40 m)	Rainy	Yes	Yes
Jan/05	12/jan	8 (0–35 m)	Rainy	Yes	Yes
	26/jan	9 (0–40 m)	Rainy	Yes	Yes
Feb/05	16/feb	9 (0–40 m)	Rainy	Yes	Yes
Mar/05	22/mar	9 (0–40 m)	Rainy	Yes	Yes
Apr/05	13/apr	9 (0–40 m)	Rainy	Yes	Yes
May/05	18/may	9 (0–40 m)	Dry	Yes	Yes
Jun/05	09/jun	9 (0–40 m)	Dry	Yes	Yes
Jul/05	05/jul	9 (0–40 m)	Dry	Yes	Yes
Aug/05	04/aug	9 (0–40 m)	Dry	Yes	No
	16/aug	9 (0–40 m)	Dry	Yes	Yes
	31/aug	8 (0–35 m)	Dry	Yes	No
Sep/05	15/sep	8 (0–35 m)	Dry	Yes	Yes
Oct/05	19/oct	8 (0–35 m)	Dry	Yes	No
Nov/05	09/nov	8 (0–35 m)	Rainy	Yes	No
	23/nov	8 (0–35 m)	Rainy	Yes	No
Dec/05	06/dec	8 (0–35 m)	Rainy	Yes	No
	21/dec	9 (0–40 m)	Rainy	Yes	No
Jan/06	04/jan	9 (0–40 m)	Rainy	Yes	No
	18/jan	9 (0–40 m)	Rainy	Yes	No
Feb/06	15/feb	8 (0–35 m)	Rainy	Yes	No
Mar/06	15/mar	8 (0–35 m)	Rainy	Yes	No
Apr/06	19/apr	8 (0–35 m)	Rainy	Yes	No
May/06	17/may	8 (0–35 m)	Dry	Yes	No
Jun/06	14/jun	8 (0–35 m)	Dry	Yes	Yes
	28/jun	8 (0–35 m)	Dry	Yes	No
Jul/06	12/jul	8 (0–35 m)	Dry	Yes	Yes
	26/jul	8 (0–35 m)	Dry	Yes	No
Aug/06	09/aug	8 (0–35 m)	Dry	Yes	Yes
	31/aug	8 (0–35 m)	Dry	Yes	No
Sep/06	21/sep	8 (0–35 m)	Dry	Yes	Yes
Oct/06	19/oct	8 (0–35 m)	Dry	Yes	Yes
Nov/06	22/nov	8 (0–35 m)	Rainy	Yes	Yes
Dec/06	07/dec	8 (0–35 m)	Rainy	Yes	No
	19/dec	8 (0–35 m)	Rainy	Yes	Yes

Considering all sampling campaigns and all sampled depths throughout the water column, a total of 317 zooplankton samples were collected and analyzed.

Samples were immediately fixed with buffered formalin (final concentration of 4%) in glass bottles. Following specialized literature (Segers, 2007, 2008) and

using an optical microscope (Olympus BX-51, Japan) and a Sedgewick–Rafter counting chamber, specimens of *K. bostoniensis* were identified and counted (ind. m⁻³) at each sampled depth.

Phytoplankton samples were collected as raw water samples and immediately fixed with acidified Lugol's solution. Sampling was conducted during two periods: from November 2004 to September 2005 and from June to December 2006. Phytoplankton, including cyanobacteria, were enumerated in random fields (Uehlinger, 1964) using the sedimentation technique (Utermöhl, 1958) under an inverted microscope (Zeiss Oberkochen, Axiovert 10, Germany). Units (cells, colonies, and filaments) were counted to a minimum of 100 individuals of the most abundant species at 400× magnification. Phytoplankton organisms were classified into the major taxonomic groups (Cyanobacteria, Chlorophyta, Chrysophyta, Bacillariophyta, Zygnematophyta, Cryptophyta, and Dinophyta). Within each taxonomic group, organisms were assigned to size classes according to their individual dimensions, considering picoplankton (0.2–2.0 µm), nanoplankton (2.0–20.0 µm), and microplankton (20.0–200 µm).

Physical and chemical variables were measured at each depth using a multiparameter probe (YSI 6600V2-4, USA), including conductivity, dissolved oxygen, and water temperature. Water samples collected at each depth were analyzed for total phosphorus, nitrate, ammonium, silica, sulfate, and orthophosphate. Nitrate, ammonium, silica, and sulfate were analyzed by ion chromatography. Replicate unfiltered samples were collected for alkalinity analysis (Gran titration) and total phosphorus (TP) determination using the ascorbic acid method following persulfate digestion in an autoclave (APHA, 2005).

Statistical analyses

To organize the data and perform the analyses, phytoplankton groups were categorized into size classes based on the densities of individual taxa, resulting in two broad categories: total nano-phytoplankton and total micro-phytoplankton. No taxa were recorded for the picophytoplankton size class. Cyanobacteria of all size classes were combined into a single group (total cyanobacteria) and analyzed separately from the other phytoplankton groups, due to their distinct ecological

dynamics in the Ribeirão das Lajes Reservoir (Recknagel et al., 2015).

The Shapiro–Wilk test was used to assess normality, indicating that all variables were non-normally distributed ($P < 0.05$). Accordingly, nonparametric tests were used for comparative analyses. The significance level adopted for all statistical tests was set at 95% ($\alpha = 0.05$). Initially, the Mann–Whitney test was used to assess whether the densities of *Kellicottia bostoniensis*, nutrients, and phytoplankton group densities varied significantly between dry (May to October) and rainy (November to April) seasons.

Physical and chemical variables were also analyzed to characterize both seasons. The vertical variation in the density of *K. bostoniensis* phytoplankton groups and physical and chemical variables along the water column were analyzed using the Kruskal–Wallis test followed by Dunn's post hoc test with the Bonferroni correction, with depth as the explanatory factor. A vertical profile representation of the reservoir was also developed using dissolved oxygen concentration and *K. bostoniensis* density for the years 2005 and 2006. The Surfer software (Golden Software LLC, 2023) was used to perform this representation, employing the Kriging interpolation method to smooth and grid the data across the depth-time matrix.

Additionally, to test Hypothesis 1, a linear mixed-effects model (LMM) was performed, with *K. bostoniensis* density as the response variable. Total phosphorus, ammonium, and depth were included as predictor variables. Depth was treated as a continuous variable, and potential nonlinear relationships were modeled using a second-order polynomial term. Seasonality was included as a fixed effect, and its interaction with depth was also evaluated. Sampling date was incorporated as a random effect to account for the non-independence of samples collected on the same day.

For Hypothesis 2, a linear mixed-effects model (LMM) was fitted with *K. bostoniensis* density as the response variable. Total cyanobacteria, total nanophytoplankton, and total micro-phytoplankton densities, along with seasonality, were included as fixed predictors. Sampling date was incorporated as a random effect to account for temporal non-independence among samples. Only samples collected on dates when both phytoplankton and zooplankton were sampled simultaneously were included in this model.

Hypothesis 3 was tested using a linear mixed-effects model (LMM), with *K. bostoniensis* density as the response variable. Dissolved oxygen and depth were included as fixed predictors, with depth treated as a continuous variable and modeled using a second-order polynomial term to account for potential non-linear patterns along the water column. Sampling date was incorporated as a random effect to control temporal non-independence among samples.

The densities of *Kellicottia bostoniensis*, total nano-phytoplankton, total micro-phytoplankton, and total cyanobacteria were log (x+1)-transformed to reduce asymmetric conditions and deal with zeros, and physical and chemical variables were standardized using the scale function in R Studio. All variables were tested with VIF before the fit model, and no evidence of multicollinearity was detected ($VIF < 3-H1$ e $H2$; $VIF \leq 5-H3$). Model assumptions were evaluated through visual inspection of diagnostic plots. Residual normality was assessed using normal Q-Q plots (qqnorm and qqline), and homoscedasticity was assessed by plotting residuals against fitted values. Diagnostic plots indicated no substantial violations of model assumptions, with residuals showing approximately constant variance and only slight deviations from normality at the distribution tails, which can be supported by LMMs (Schielzeth et al., 2020). The residual autocorrelation for temporal scale was assessed using the autocorrelation function (ACF) (Zuur et al., 2010).

All analyses were conducted in the R environment (version 4.5.2), and LMMs were fitted with the lmer function from the package lme4 with sampling date incorporated as a random effect to account for the non-independence of samples collected on the same day. Also, we use the package's stats (Shapiro-Wilk, Kruskal-Wallis, Mann-Whitney, qqnorm, qqline, scale, acf) and performance (r2) (Lefcheck, 2016; Hartig, 2024; R Core Team, 2025).

Results

Seasonal characterization

Kellicottia bostoniensis density was significantly higher ($W = 1640$, $P < 0.001$) during the dry season (May to October), with a mean of 3506.8 ind. m^{-3} , compared to the rainy season (November to April),

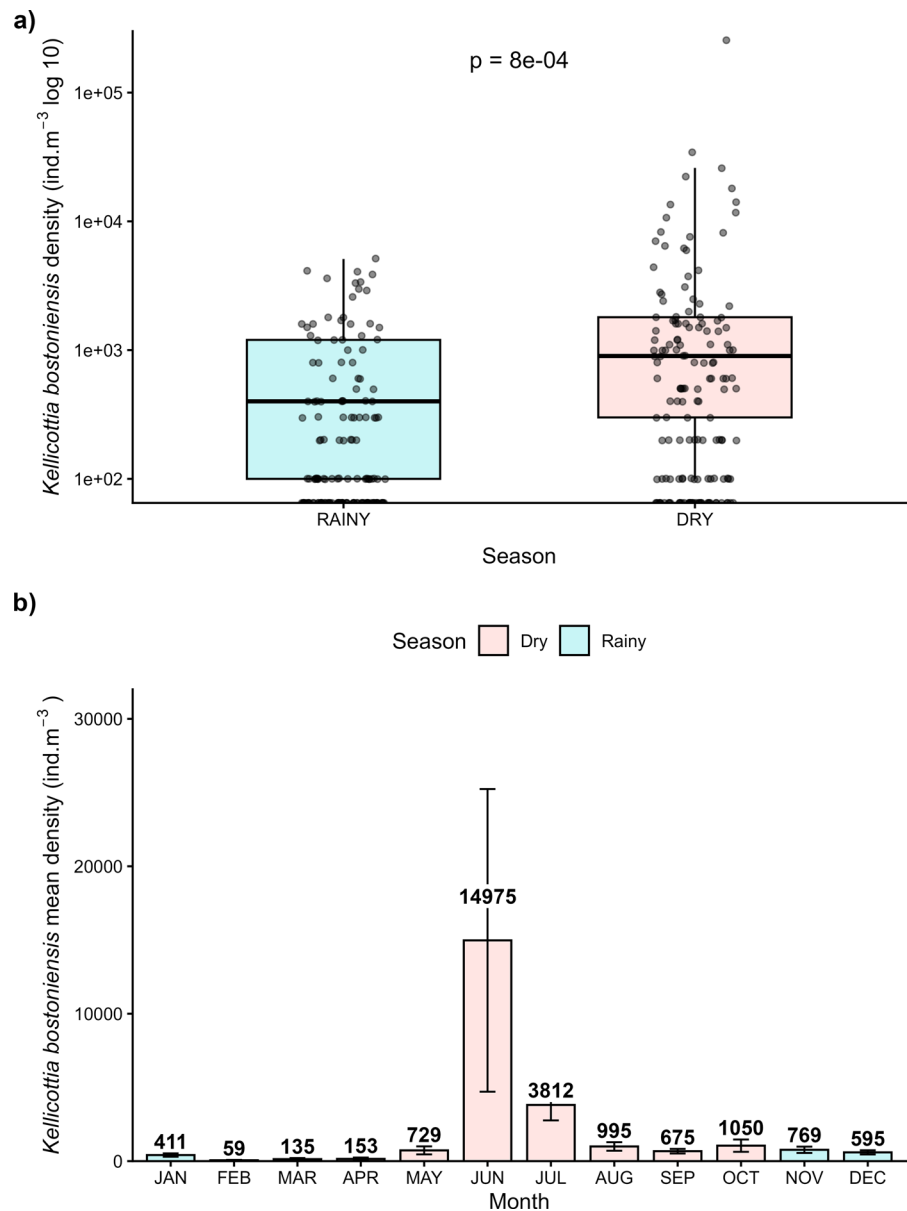
with a mean of 483 ind. m^{-3} (Fig. 2a, Table S1). During the dry season, the highest mean density was observed in June (14,975 ind. m^{-3}), followed by July (3812 ind. m^{-3}). Conversely, lower mean densities were recorded in February (59 ind. m^{-3}), March (135 ind. m^{-3}), and April (153 ind. m^{-3}), during the rainy season (Fig. 2b).

Except for water temperature, nitrate, and total phosphorus, physical and chemical variables differed significantly between the dry and rainy seasons, with higher values during the dry season. The mean dissolved oxygen concentration was 4 mg l^{-1} in the dry season, compared to 2.4 mg l^{-1} in the rainy season ($W = 1659$, $P < 0.001$). The mean of electrical conductivity was 29.8 $\mu S\ cm^{-1}$ in the dry season and 29.1 $\mu S\ cm^{-1}$ in the rainy season ($W = 9980$, $P = 0.001$). For nutrients, orthophosphate mean was 6.5 $\mu g\ l^{-1}$ in the dry season and 5.7 $\mu g\ l^{-1}$ in the rainy season ($W = 1507$, $P = 0.002$). In comparison, ammonium varied from 541.3 $\mu g\ l^{-1}$ in the dry season to 408.6 $\mu g\ l^{-1}$ in the rainy season ($W = 1528$, $P < 0.001$). Nitrate did not differ significantly, with 54.3 $\mu g\ l^{-1}$ in the dry season and 53.9 $\mu g\ l^{-1}$ in the rainy season (Table S2).

Among phytoplankton groups, total nano-phytoplankton was significantly higher ($W = 5433.5$, $P < 0.001$) in dry season, with a mean of 1396.97 ind. ml^{-1} . No significant differences occurred in total cyanobacteria density between seasons ($W = 3213$, $P = 0.071$). Still, following an opposite trend, the group showed higher mean densities in rainy seasons (1889.7 ind. ml^{-1}), than in dry seasons (1296.5 ind. ml^{-1}) (Table S3).

The LMM analysis (Table 2) revealed that ammonium and seasonality were the main drivers of *K. bostoniensis* density. Among the fixed effects, ammonium showed a significant negative effect ($P = 0.019$), indicating a decrease in density with increasing concentrations. In contrast, total phosphorus had no significant effect on species density ($P = 0.766$). Depth exhibited a significant nonlinear pattern, with the second-order polynomial term being strongly negative ($P < 0.001$), suggesting variation in density along the water column, with density increasing at intermediate depths and decreasing at the extremes. Seasonality also had a significant effect, with lower densities observed during the rainy season ($P = 0.012$). The interaction between depth and seasonality was not statistically significant either for the first-order term

Fig. 2 Seasonal **a** and monthly **b** variation in the density of *Kellicottia bostoniensis* in Ribeirão das Lajes Reservoir. Seasonal density is expressed in individuals per cubic meter (ind. m^{-3}), and the y-axis is shown on a log10 scale. Monthly density is expressed by the mean of individuals (ind. m^{-3}) in each month



($P = 0.481$) or for the second-order term ($P = 0.072$), indicating that the vertical distribution pattern of density does not vary with season. These results suggest that seasonality affects only the overall magnitude of abundance, with higher densities observed during the dry season, rather than altering the shape of the vertical profile (Fig. 3). The R^2 for the conditional model represents 49.30% of the total explained variance. However, marginal effects account for 27.80% of this result, suggesting that fixed factors carry greater weight than random factors.

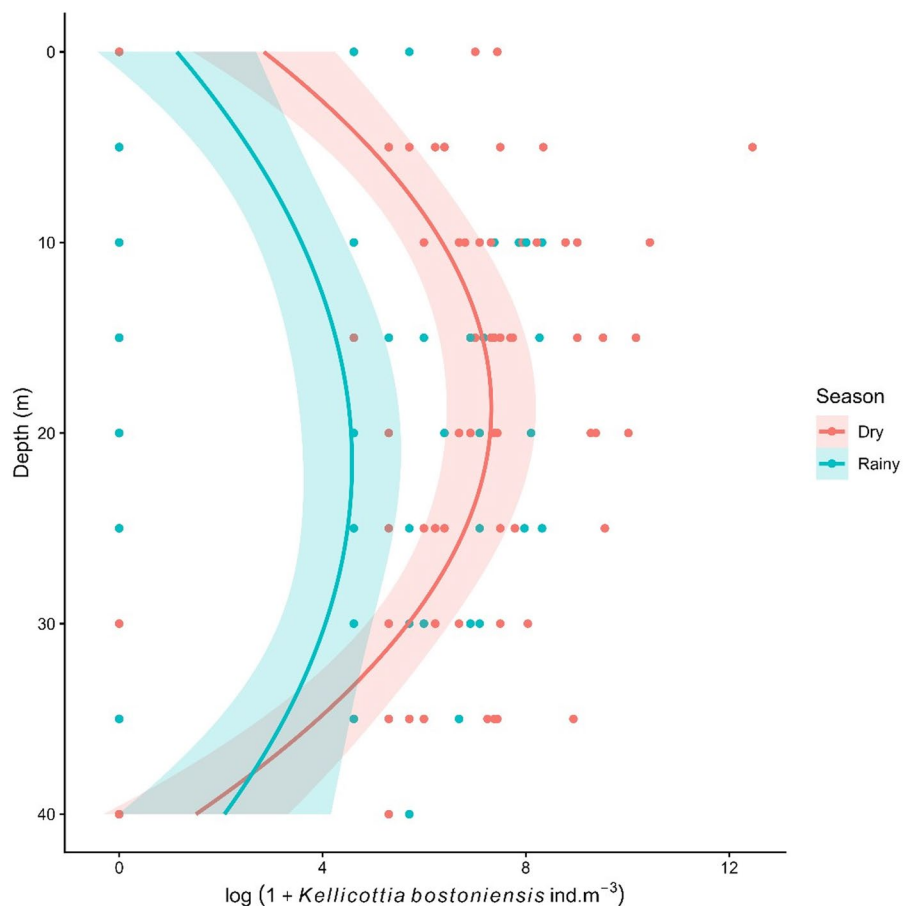
LMM analysis (Table 3) indicated that phytoplankton groups and seasonality influenced *K. bostoniensis* density. Among the fixed effects, total nano-phytoplankton showed a significant positive effect ($P = 0.047$), supporting the hypothesis that increases in density during the mixing period are associated with higher nano-phytoplankton availability. In contrast, total cyanobacteria exhibited a significant negative effect ($P = 0.034$), indicating a reduction in *K. bostoniensis* density with increasing cyanobacterial abundance. Total micro-phytoplankton showed no

Table 2 Results of the linear mixed-effects model (LMM) evaluating the effects of total phosphorus (P_{total}), ammonium, depth, and seasonality on *Kellicottia bostoniensis* density

Fixed effects	Estimate	Std. Error	df	<i>t</i> value	<i>P</i> -value	Significance
(Intercept)	5.6856	0.5646	16.3554	10.069	< 0.001	***
Total phosphorus	0.0676	0.2270	164.0396	0.298	0.766	
Ammonium	-0.6970	0.2943	127.6143	-2.369	0.019	*
poly(depth, 2)1	3.8935	3.7184	161.3954	1.047	0.297	
poly(depth, 2)2	-20.3714	3.4986	154.9654	-5.823	< 0.001	***
season (rainy)	-2.5437	0.8986	15.7737	-2.831	0.012	*
poly(depth, 2)1 × season (rainy)	3.6988	5.2333	157.5700	0.707	0.481	
poly(depth, 2)2 × season (rainy)	9.1631	5.0623	152.4746	1.810	0.072	

Estimates, standard errors (Std. Error), degrees of freedom (df), *t*-values, *P*-values, and significance codes are presented. Significance codes: ****P* < 0.001; ***P* < 0.01; **P* < 0.05; *P* < 0.1

Fig. 3 Vertical distribution of *Kellicottia bostoniensis* density across depth (0–40 m) during dry and rainy seasons. Points represent observed values of the log-transformed density [$\log(x+1)$], while solid lines indicate fitted smooth trends for each season (dry in red and rainy in blue). Shaded areas represent 95% confidence intervals around the fitted curves. Higher overall densities were observed during the dry season, whereas both seasons showed nonlinear vertical patterns with peak values at intermediate depths



significant effect ($P = 0.938$). Seasonality was marginally significant in the model ($P = 0.052$), with a trend toward higher densities during the dry season; this pattern is consistent with the Mann–Whitney test results, which indicated significantly higher densities

in the dry season. The R^2 for the conditional model accounts for 32.10% of the total explained variance, although marginal effects account for 13.50% of this result, increasing the weight of random factors in this case.

Table 3 Results of the linear mixed-effects model (LMM) evaluating the effects of total cyanobacteria, total nano-phytoplankton, total micro-phytoplankton, and seasonality on *Kellicottia bostoniensis* density

Fixed effects	Estimate	Std. Error	df	t value	P-value	Significance
(Intercept)	5.233011	0.923380	85.088121	5.667	< 0.001	***
Total cyanobacteria	-0.225635	0.105512	169.622477	-2.138	0.034	*
Total nano-phytoplankton	0.246703	0.123142	169.998723	2.003	0.047	*
Total micro-phytoplankton	-0.007264	0.092949	169.066966	-0.078	0.938	
Season (rainy)	-1.821614	0.866338	15.897777	-2.103	0.051	

Estimates, standard errors (Std. Error), degrees of freedom (df), t-values, p-values, and significance codes are presented. Significance codes: *** $P < 0.001$; ** $P < 0.01$; * $P < 0.05$; $P < 0.1$

Vertical profile characterization

Kellicottia bostoniensis was present and varied significantly across all sampling depths ($P < 0.05$). Dunn's post hoc test indicated a consistent vertical distribution pattern, with significantly lower densities at the surface (100 ind. m^{-3} at 0 m) compared to 10, 15, and 20 m ($P \text{ adj} < 0.001$). The highest mean at intermediate depths (5–25 m) ranged, respectively, from 7113 to 1079 ind. m^{-3} . A significant decrease was observed in deeper layers, particularly at 40 m, where the lowest mean density was recorded (53.8 ind. m^{-3}) (15 m vs. 40 m: $Z = 4.59$; $P \text{ adj} < 0.001$) (Fig. 4). Further values are provided in Table S4.

The reservoir showed clear vertical stratification of the water column for about eight-to-nine months (from October to May–June), with temperature ($H = 247.0$, $P < 0.001$), dissolved oxygen ($H = 208.0$, $P < 0.001$), conductivity ($H = 86.4$, $P < 0.001$), orthophosphate ($H = 15.6$, $P = 0.048$), total phosphorus ($H = 21.2$, $P = 0.006$), nitrate ($H = 55.3$, $P < 0.001$), and ammonium ($H = 36.1$, $P < 0.001$) varying significantly across depth strata.

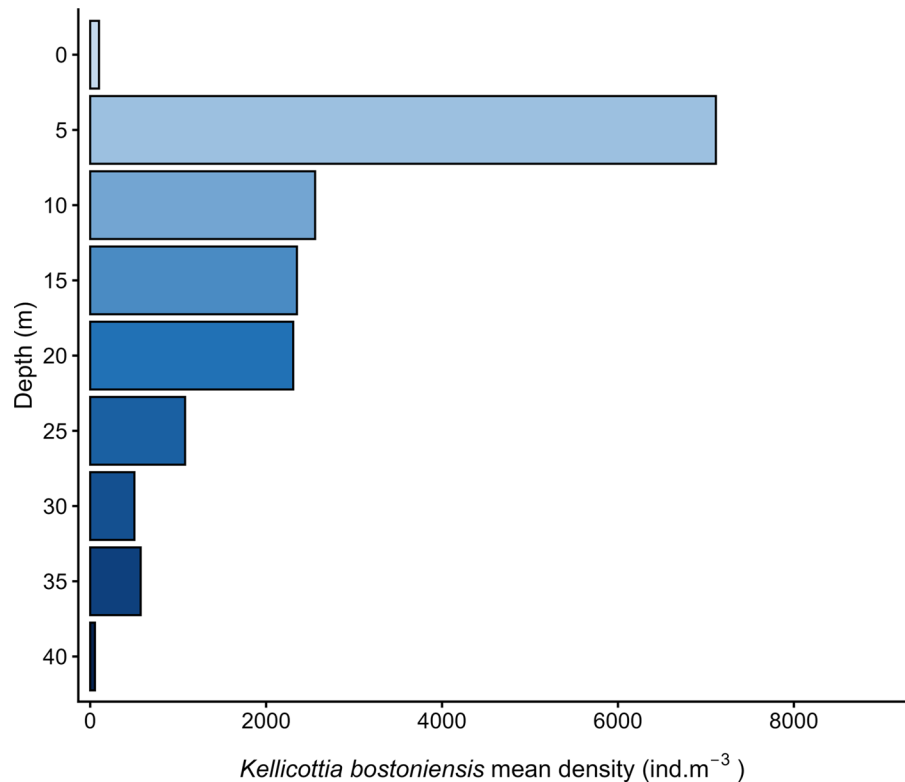
Thermal stratification was characterized by higher temperatures at the reservoir surface (0 m), with a mean of 25.7 °C, defining the epilimnion. Temperature decreased from 10–15 m, with mean values of 22.7 °C at 10 m and 21.7 °C at 15 m, indicating the onset of the metalimnion. Between 20 and 40 m, the lowest temperatures were recorded (19.5–22.2 °C), characterizing the hypolimnion. Dissolved oxygen ranged from 5.3 to 11.7 mg l^{-1} at the surface (mean = 8.2 mg l^{-1}) but decreased sharply with depth, reaching mean values of 4.9 mg l^{-1} at 10 m and 1.9 mg l^{-1} at 15 m. Below 25 m, concentrations were very low (0.7 mg l^{-1}), reaching anoxic conditions at 40 m (0–0.3 mg l^{-1}). In contrast, conductivity averaged

27.1 $\mu S \text{ cm}^{-1}$ at the surface, decreased between 10 and 15 m (minimum mean of 25.6 $\mu S \text{ cm}^{-1}$ at 10 m), and increased again in deeper layers, reaching 44.9 $\mu S \text{ cm}^{-1}$ at 40 m (Table S5).

Phytoplankton showed significant variation across depths for total cyanobacteria (Kruskal–Wallis, $\chi^2 = 48.40$, $P < 0.001$) and total nano-phytoplankton ($\chi^2 = 25.88$, $P = 0.001$). Pairwise comparisons revealed that total cyanobacteria differed significantly between the surface (0 m) and all deeper layers from 15 to 40 m, as well as between shallow (5 m) and deeper (20–40 m) layers. For total nano-phytoplankton, significant differences were mainly observed between the surface and deeper layers (0 m vs. 35–40 m) and between 10 and 40 m. No significant variation and pairwise differences were detected for total micro-phytoplankton after Bonferroni correction. Total cyanobacteria presented higher means at shallower depths, with a peak at 0 m (3885.3 ind. ml^{-1}); lower mean values were registered at 30 m (542.7 ind. ml^{-1}). At 40 m, the mean was 644.8 ind. ml^{-1} . Total nano-phytoplankton were more abundant at the surface (0 m), with 2101.7 ind. l^{-1} of mean, and lower mean densities were registered at the other extremity (40 m), with the value of 123 ind. ml^{-1} . The same was observed for the total micro-phytoplankton group, higher mean densities at 0 m (420.3 ind. ml^{-1}), and lower at 40 m (78.7 ind. ml^{-1}) (Table S6).

The vertical depth profile revealed that persistent stratification occurred throughout most of 2005 and 2006 (Fig. 5). Dissolved oxygen (DO) concentration varied across water-column layers, with lower values below 15 m, indicating an anoxic hypolimnion for most part of the year. However, this condition changes with the beginning of the mixing of the reservoir at the end of autumn (May) and in winter, when higher oxygen content is found below 15 m. *Kellicottia*

Fig. 4 Mean values of *Kellicottia bostoniensis* densities in all sampling depths during the period of the study



bostoniensis was present at almost all depths except at the end of the stratification period (May 2005 and March and April 2006).

The third LMM analysis (Table 4) indicated that dissolved oxygen and depth significantly influenced the density of *K. bostoniensis* along the water column. Dissolved oxygen showed a significant positive effect ($P = 0.025$) in the species density, indicating higher densities under more oxygenated conditions, such intermediate depths and during mixing conditions. However, under low-oxygen conditions, the densities are much lower in the deepest hypolimnetic layers at the end of the stratification period (Fig. 5). Depth exhibited a strong nonlinear relationship, with both first- and second-order polynomial terms being significant ($P = 0.007$ and $P < 0.001$, respectively), indicating a structured vertical distribution pattern. Despite the positive effect of dissolved oxygen, *K. bostoniensis* density was not high at the surface, reflecting the nonlinear variation along the depth gradient. In this way, the R^2 for the conditional model accounts for 41.20% of the total explained variance, although marginal effects account for 18.40% of this result, suggesting grater weights for random factors.

Discussion

Aiming to determine how water-column conditions in a deep tropical reservoir may favor or hinder the presence of the non-native rotifer *Kellicottia bostoniensis*, we found that the species occurred across the vertical profile at several depths, under both stratification and mixing conditions throughout the two studied years. The reported presence and persistence of *K. bostoniensis* in the Lajes Reservoir since 2001 (Lopes et al., 2018; Branco et al., 2023) indicate that the species has progressed beyond the introduction stage and become successfully established in the reservoir. Such long-term persistence suggests that the environmental conditions of Ribeirão das Lajes consistently support viable populations and attests to success in maintaining its non-native populations in the reservoir.

Understanding the persistence and integration of a non-native species into an invaded area over a long timescale has received less attention (Puth and Post, 2005; Macallum et al., 2014). Thus, it can be challenging for aquatic microscopic organisms, as physical, chemical, and biological drivers act simultaneously on them. Deep tropical reservoirs are perfect for

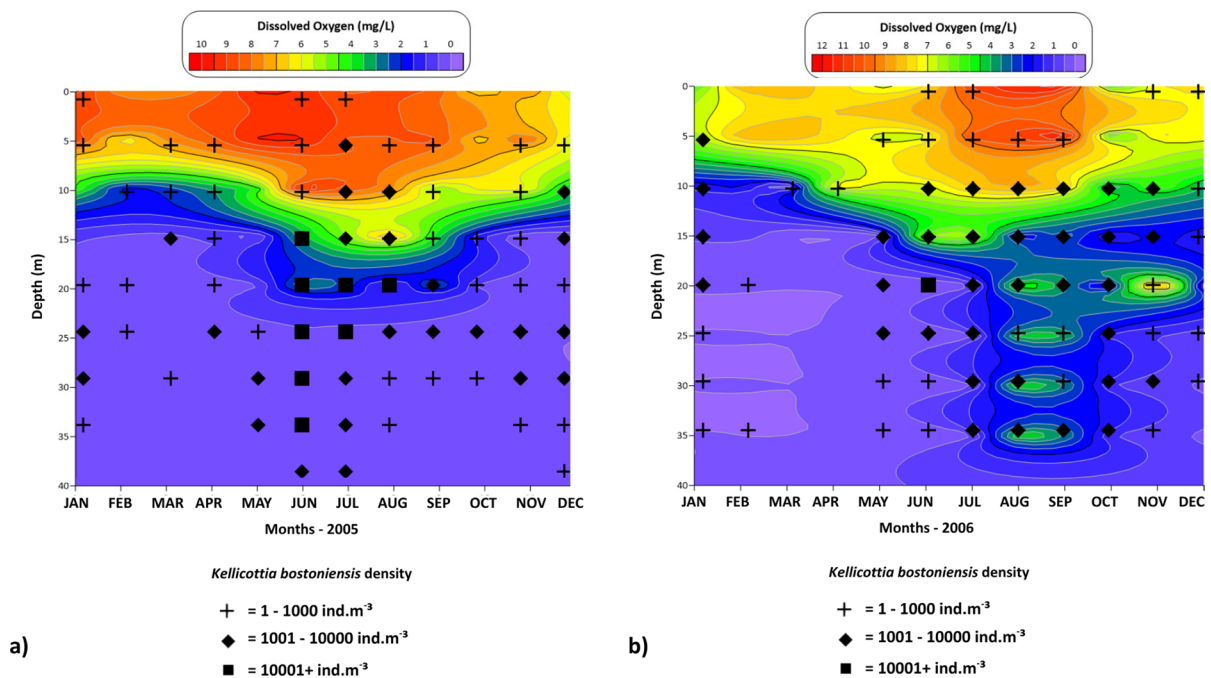


Fig. 5 Profile of oxygen concentration and density of *Kellicottia bostoniensis* at all sampling depths during the months of 2005 (a) and 2006 (b)

Table 4 Results of the linear mixed-effects model (LMM) evaluating the effects of dissolved oxygen and depth on *Kellicottia bostoniensis* density

Fixed effects	Estimate	Std. Error	df	t value	P-value	Significance
(Intercept)	4.5741	0.4283	15.8921	10.679	< 0.001	***
Dissolved oxygen	1.0403	0.4604	168.5232	2.259	0.025	*
Depth (poly 1)	14.9661	5.4613	170.9994	2.740	0.007	**
Depth (poly 2)	-21.4106	3.2361	167.7158	-6.616	<0.001	***

Estimates, standard errors (Std. Error), degrees of freedom (df), t-values, p-values, and significance codes are presented. Significance codes: *** $P < 0.001$; ** $P < 0.01$; * $P < 0.05$; . $P < 0.1$

examining invasion persistence. Because of their pronounced vertical gradients and predictable seasonal thermal dynamics, they generate a mosaic of environmental conditions across the water column. Rather than representing just one massive, identical habitat, these systems contain multiple ecological niches that may facilitate the persistence of non-native planktonic species capable of exploiting spatial and temporal environmental heterogeneity.

To our knowledge, the present study is the first in a tropical reservoir to track, for such a long time, the occurrence and densities of *K. bostoniensis* across the entire water column. In our study, we collected 317

samples with a fortnightly to monthly frequency. As expected by our first hypothesis, the *K. bostoniensis* population was influenced by seasonality, as it occurred and exhibited higher densities in the dry season at all depths. Consistent with previous studies in the Ribeirão das Lajes Reservoir (Branco et al., 2023), *K. bostoniensis* showed a marked increase in density during the dry season, likely due to water-column mixing.

In Ribeirão das Lajes Reservoir, mixing conditions are known to promote marked changes in the physical and chemical structure of the water column, including increases in nutrient concentrations and chlorophyll-*a*

throughout the profile (Guarino et al., 2005; Branco et al., 2009). Column mixing in Ribeirão das Lajes Reservoir is considered a key event, as it promotes the destratification of a water column that can remain stratified for up to nine months (Rocha et al., 2019). This process results in the mixing of a nutrient-poor, oxygenated epilimnion with a nutrient-rich, anoxic hypolimnion (Guarino et al., 2005), leading to increased ammonium concentrations that can be partially oxidized to nitrate following oxygen redistribution (Laiz et al., 1994).

However, the relatively high ammonium concentrations compared to nitrate suggest that nitrogen availability may be influenced by reduced forms associated with internal nutrient recycling processes (Heisler et al., 2008), possibly reflecting internal recycling rather than external nutrient inputs. Accumulation of organic matter and ammonium in the bottom layer of stratified reservoir and increase during the mixing period besides found in Ribeirão das Lajes (Guarino et al., 2005; Branco et al., 2009) was already reported for other Brazilian reservoirs (Tundisi, 1994). Because ammonium accumulation in freshwater systems is often linked to oxygen depletion and hypolimnetic processes, including reduced nitrification and sediment nutrient release (Beutel, 2001, 2006), its relationship with *K. bostoniensis* may also reflect the broader environmental conditions under which it accumulates rather than a direct effect of ammonium concentration itself.

In particular, the negative association of *K. bostoniensis* density with increased ammonium concentration indicates that the species may respond more strongly to the broader ecological conditions associated with the mixing regime than to nutrient concentration alone. Instead of responding to nutrient concentrations per se, the species is more strongly associated with changes in the phytoplankton community structure. In our study, the response of *K. bostoniensis* was not directly explained by a general increase in nutrient availability, indicating that nutrients do not act as primary drivers of its population dynamics.

From an invasion ecology perspective, seasonal mixing may represent more than a limnological process affecting resource availability. By temporarily increasing habitat connectivity and redistributing environmental conditions throughout the water column, mixing may also enhance the performance of

already established populations. Our findings support this mechanism: The elevated densities observed during the dry season suggest that mixing does not facilitate initial colonization but rather promotes conditions that favor population growth of an already established non-native species.

Smaller phytoplankton fractions, particularly nano-phytoplankton (<20 µm), showed a positive association with *K. bostoniensis*, whereas cyanobacteria were negatively related to its density. This pattern suggests that, although mixing enhances nutrient availability, it may favor phytoplankton groups that are more suitable as food resources while simultaneously reducing the dominance of less favorable groups (Stockenreiter et al., 2021; Bandara et al., 2025). Periods of stratified conditions lead to heterogeneous phytoplankton distribution throughout the epilimnion, metalimnion, and hypolimnion (Naselli-Flores et al., 2007; Becker et al., 2008). In stratified reservoirs, low turbulence and resource gradients often result in phytoplankton maxima in specific layers of the water column, reflecting distinct ecological strategies linked to the simultaneous availability of light and nutrients (Mellard et al., 2011; Recknagel et al., 2015). Furthermore, the stability of the metalimnion may promote spatial separation of phytoplankton groups and increase vertical community heterogeneity, as stratification creates marked physical and chemical differences among layers (T-Krasznai et al., 2024).

Our results therefore support the second hypothesis. The LMM results showed a significant positive relationship between *K. bostoniensis* and nano-phytoplankton, suggesting that this size class may represent an important food resource for the species under mixing conditions. In contrast, cyanobacteria exhibited a significant negative association with *K. bostoniensis* density. These findings reinforce that the response of *K. bostoniensis* is not driven by total phytoplankton abundance, but rather by changes in community composition and size structure. In this context, nano-phytoplankton appears to play a positive role, whereas cyanobacterial dominance may not favor the species, potentially due to ecological characteristics of these phytoplankton groups (Trommer et al., 2019; Ling et al., 2025). The observed higher nano-phytoplankton availability during dry seasons and mixing may have contributed to more favorable conditions for *K. bostoniensis*, potentially through

decreased competition for food and improved trophic conditions.

Variations in phytoplankton density and composition along the vertical profile are directly associated with zooplankton distribution, as gradients of food, oxygen, and temperature determine habitat suitability across depths (Sanful et al., 2017). In addition, local maxima of phytoplankton density at specific depths may favor aggregations of herbivorous zooplankton, which tend to exploit regions with greater availability of suitable food particles (Greer et al., 2013). Since *K. bostoniensis* is primarily a microfilter-feeding rotifer but may also ingest fine detritus, bacteria and organic aggregates, it can be considered polyphagous (Gilbert, 2022). Moreover, species of *Kellicottia* exhibit greater selectivity for particle size ($<20\ \mu\text{m}$) compared to other genera within the family Brachionidae (e.g., *Brachionus*) (Gilbert, 2022). These omnivorous characteristics may help explain the higher densities of *Kellicottia* found in Ribeirão das Lajes reservoir throughout the year at a depth of 5 m, where primary production by phytoplankton and bacteria should be higher.

The third hypothesis, which proposed *K. bostoniensis* as a euryecious species capable of occurring across the entire water column, even under low-oxygen conditions, was partially supported by our results. In stratified systems, hypoxia in the hypolimnion can restrict habitat availability and alter zooplankton biomass and community composition (Doubek et al., 2019; Karpowicz et al., 2020). Nevertheless, rotifers generally exhibit greater tolerance to oxygen depletion than microcrustaceans (Snell et al., 2019), although many species still attain their highest densities under well-oxygenated conditions, where physiological constraints are reduced and trophic resources are more favorable (Devkota et al., 2025).

Consistent with this pattern, *K. bostoniensis* occurred throughout the water column, but its density was positively associated with dissolved oxygen, indicating a preference for more oxygenated habitats despite its ability to persist under lower oxygen concentrations. Dissolved oxygen therefore appears to act as an important environmental filter shaping its vertical distribution, while depth exerted a significant nonlinear effect, with peak densities occurring at intermediate layers and lower densities near both the surface and the bottom. Higher oxygen concentrations during the mixing period likely contributed to

this distribution by expanding suitable habitat across the water column.

The occupation of multiple depth strata suggests that *K. bostoniensis* can exploit distinct ecological niches within the reservoir, a characteristic commonly associated with successful non-native aquatic species (Shea & Chesson, 2002). However, dissolved oxygen and food availability alone are unlikely to fully explain the aggregation of individuals at intermediate depths. Vertical habitat selection in zooplankton reflects a balance among multiple ecological trade-offs, including resource acquisition, metabolic costs, predation risk, and physical gradients (Zaret & Suffern, 1976; Lampert, 2005; Williamson et al., 2011). In tropical reservoirs, the metalimnion may provide this balance by combining suitable oxygen concentrations and food availability while reducing exposure to visually oriented predators, excessive irradiance, and the greater physical instability characteristic of surface waters (Wetzel, 2001; Becker et al., 2008). Although these mechanisms were not directly evaluated here, they provide plausible complementary explanations for the vertical aggregation of *K. bostoniensis*.

Our results indicate that the long-term persistence of *K. bostoniensis* in Ribeirão das Lajes Reservoir is associated not only with seasonal changes in limnological conditions but also with its ability to exploit the predictable spatial and temporal heterogeneity of a deep tropical reservoir. The combination of broad environmental tolerance, vertical habitat exploit, and responses to seasonal mixing likely contributes to the successful establishment and persistence of this non-native rotifer, illustrating how reservoir characteristics can facilitate the ecological integration of invasive zooplankton.

Conclusion

Overall, the dynamics of the non-native rotifer *Kellicottia bostoniensis* in the Ribeirão das Lajes Reservoir were primarily driven by seasonal mixing during dry seasons, when the species achieved its highest densities. Regarding food resources, our findings indicate that phytoplankton composition and size structure, specifically the positive relationship with nano-phytoplankton and the negative association with cyanobacteria, rather than total phytoplankton

abundance, are key drivers of its population dynamics. Vertically, *K. bostoniensis* distribution exhibited a significant nonlinear pattern, with peak densities restricted to intermediate depths, reflecting a balance between dissolved oxygen requirements and environmental trade-offs across the water column. From an invasion ecology perspective, the ability to exploit this predictable temporal and vertical heterogeneity facilitates the species' integration and long-term persistence in deep tropical systems. Ultimately, this structural variability highlights the critical importance of depth-integrated sampling to properly assess the abundance and establishment success of invasive zooplankton in stratified aquatic ecosystems.

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Authors contributions All authors contributed significantly toward the conception of this manuscript. Christina Wyss Castelo Branco did conceptualization; Lucas Cesar Santos Cabral, Raphael Ferreira Correa, Karen Helena Costa Santos, Vanessa Guimarães Lopes, Vera Huszar, Betina Kozlowsky-Suzuki, and Christina Wyss Castelo Branco gave the methodology; Lucas Cesar Santos Cabral, Raphael Ferreira Correa, Karen Helena Costa Santos, and Christina Wyss Castelo Branco performed formal analysis and investigation; Lucas Cesar Santos Cabral did writing—original draft preparation; Lucas Cesar Santos Cabral, Raphael Ferreira Correa, Karen Helena Costa Santos, Vanessa Guimarães Lopes, Vera Huszar, Adriana Lamanna Puga, Betina Kozlowsky-Suzuki, and Christina Wyss Castelo Branco done writing—review and editing; Lucas Cesar Santos Cabral and Christina Wyss Castelo Branco supervised the study.

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Data availability The datasets generated during the current review are available from the corresponding author on reasonable request.

Code availability Not applicable.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval Not applicable.

Consent to participate Not applicable.

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